ACTIVITY 6: Testing the Account Class with Exceptions

Objective

Write a test class that handles exceptions thrown by the enhanced `Account` class.

Requirements

Create a test class `TestAccountExceptions.java` that:

1. Tests all scenarios with proper try-catch blocks
2. Displays informative messages when exceptions occur
3. Shows both success and failure cases
4. Demonstrates proper exception handling

Expected Output

When you run your test class, the output should look like this:

```
text
============================================================
  ACCOUNT TEST WITH EXCEPTIONS
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>>> Test 1: Valid Account Creation
SUCCESS: Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

>>> Test 2: Invalid Age (under 18)
EXCEPTION: Customer must be at least 18 years old. Provided: 16

>>> Test 3: Invalid Account Type
EXCEPTION: Account type must be 'Savings' or 'Current'. Provided: Invalid

>>> Test 4: Minimum Balance on Creation
```

Creating Savings account with ₹300
EXCEPTION: Savings account requires minimum balance of ₹500.0. Provided: ₹300.0

>>> Test 5: Valid Deposit and Withdrawal
Account: Account #1005 | Alice Brown (30 yrs) | Current | ₹1000.0 | Active | PIN: No
Setting PIN 1234: SUCCESS
Depositing ₹500.0: SUCCESS
Balance after deposit: ₹1500.0
Withdrawing ₹200.0: SUCCESS
Balance after withdrawal: ₹1300.0
Account #1005 | Alice Brown (30 yrs) | Current | ₹1300.0 | Active | PIN: Yes

>>> Test 6: Invalid Deposit (Negative Amount)
Attempting to deposit ₹-100.0
EXCEPTION: Deposit amount must be positive. Provided: ₹-100.0

>>> Test 7: Insufficient Balance
Account: Account #1006 | Charlie Green (35 yrs) | Savings | ₹500.0 | Active | PIN: Yes
Attempting to withdraw ₹1000.0
EXCEPTION: Insufficient balance. Available: ₹500.0, Requested: ₹1000.0

>>> Test 8: Minimum Balance Violation
Account: Account #1007 | Diana Prince (28 yrs) | Savings | ₹1000.0 | Active | PIN: Yes
Attempting to withdraw ₹600.0
EXCEPTION: Cannot withdraw. Minimum balance of ₹500.0 required. Available after withdrawal: ₹400.0

>>> Test 9: Inactive Account Operations
Account: Account #1008 | Eve Wilson (32 yrs) | Current | ₹2000.0 | Active | PIN: No
Closing account: SUCCESS
Attempting to deposit ₹100.0 on closed account
EXCEPTION: Account is inactive. Please reopen the account or contact support.
Reopening account: SUCCESS
Depositing ₹100.0 after reopen: SUCCESS
Balance after deposit: ₹2100.0

>>> Test 10: PIN Verification
Account: Account #1009 | Frank Miller (40 yrs) | Savings | ₹1500.0 | Active | PIN: No
Setting PIN 1234: SUCCESS
Withdrawing ₹200.0 with correct PIN: SUCCESS

```
Balance: ₹1300.0
Attempting to withdraw ₹100.0 with incorrect PIN (9999)
EXCEPTION: Incorrect PIN
Attempting to withdraw ₹100.0 without PIN set
EXCEPTION: PIN not set for this account

>>> Test 11: All Accounts Summary
Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No
Account #1005 | Alice Brown (30 yrs) | Current | ₹1300.0 | Active | PIN: Yes
Account #1006 | Charlie Green (35 yrs) | Savings | ₹500.0 | Active | PIN: Yes
Account #1007 | Diana Prince (28 yrs) | Savings | ₹1000.0 | Active | PIN: Yes
Account #1008 | Eve Wilson (32 yrs) | Current | ₹2100.0 | Active | PIN: No
Account #1009 | Frank Miller (40 yrs) | Savings | ₹1300.0 | Active | PIN: Yes

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  TEST COMPLETED!
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```

Task

Write the complete `TestAccountExceptions` class in a file called `TestAccountExceptions.java`. Handle all exceptions with try-catch blocks.